A Comprehensive Review of AI-Driven NPC Intelligence and Procedural Content Generation in Video Games

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Abstract - The review presents the results of impactful articles published in the period between 2011 and 2025 and is based on high-impact publications, including the Nature Machine Intelligence, IEEE Transactions on Games, and major ACM conferences. It is a comparative study of the major methodologies, such as deep learning, reinforcement learning, evolutionary algorithms, large language models, and generative AI.With the introduction of the advanced artificial intelligence, the development of video games has undergone a fundamental change, in terms of the Non-Player Character (NPC) behavior and Procedural Content Generation (PCG). Such evolution has made it possible to produce adaptive and lifelike NPCs, as well as the production of game environments, levels and narratives dynamically and automatically. The article follows the path of the field since the early rule-based systems to the modern techniques of data, showcasing the key innovations and their recorded influence on the field.

Keywords: Non player characters (NPC), Procedural Content Generation (PCG), Artificial Intelligence, Machine Learning, Game Development, Reinforcement Learning, Deep Learning, Large Language Models.

1. INTRODUCTION

Artificial intelligence has come to be a revolutionizing technology in the video game business, leading to major innovations in two major aspects: intelligent non-player characters (NPC) and procedural content generation (PCG). PCG enables the creation of game environments, levels and narrative in a dynamic and static manner. It allows the NPCs to be life-like and the interactions to be meaningful. NPCs have now become not only challenging and contextually meaningful but also can be a human companion, thus improving the gameplay with behaviors that are being increasingly seen as adaptive and realistic.

The review presents the conclusions of powerful studies, which were published as early as 2011, and 2025, and therefore, the field is rapidly evolving. It gives a general overview of studies in NPC intelligence and PCG, presents a comparative study of methodological strategies and their findings, comments on the existing issues and future perspectives, and ends with a reflection of what researchers and practitioners can learn.

2. RELATED WORKS

This section provides a systematic analysis of 30 research papers focusing on AI-driven NPC intelligence and procedural content generation, organized chronologically to demonstrate the evolution of techniques and methodologies in this rapidly advancing field.

A. Foundational Era (2011-2016)

The foundational period established key theoretical frame- works and early practical implementations that would influence subsequent research directions.

- Yannakakis and Togelius [3] proposed the concept of experience-based procedural content generation. Player models provided foundation for player-adaptive content which is crucial for creating personalized gaming experience.
- One comprehensive initial survey by Hendrikx et al. [2] established the taxonomic

structure of procedural content generation that is widely applied nowadays. Their methodical deconstruction of the algorithms, space and the combination of AI developed a conceptual framework on the implementation of PCG across genres.

- Ponce and Padilla [17] used hierarchical reinforcement learning schemes in order to produce intelligent NPCs. Al-Emran [8] presented a synthetic description of hierarchical reinforcement learning algorithms. This piece was a theoretical basis of multi-agent systems and time abstraction that were vital in the later works of NPC behavior modeling.Hierarchical reinforcement learning provides multilevel decision making mechanisms that allow NPCs to learn complex behaviors during their interaction with environment, giving more flexibility than the traditional finite state machines.

- Feng et al. [18] used a combination of machine learning with behavior trees to develop autonomous behavior learning systems among NPC. This strategy made the NPCs able to acquire knowledge through engagement with players and adjust their behavior strategies, which contributed significantly to the development of realistic adversaries.

B. Deep Learning Revolution (2017-2020)

- The period was also characterized by the use of the original methods of deep learning in the field of game AI, which triggered the development of new methods in the intelligence of NPCs and generation of content
- Snodgrass and Onta Nuna [9] proposed a procedure to manage the content generation by constrained multi-dimensional Markov chain sampling. Their work showed that with probabilistic approaches it is possible to have fine-grained control over the generated content and remain diverse.
- The concept of PCG using machine learning was introduced early in a large-scale study by Summerville et al. [4], who introduced taxonomies to inform future studies.

- Zafar et al. [15] explored the idea of search-based procedural generation of general video game content. Their study revealed that evolutionary techniques are useful in creating content in most game genres.
- It was shown that increasing the generality of machine learning by procedural content generation could be done by Risi and Togelius [5] showed the way in which procedural content generation could be applied in order to increase the generality of machine learning. Their contribution established PCG as a possible testbed on which to develop more general AI systems and thus established principal connections between the game AI and the broader discipline of machine learning.

C. Modern AI Integration (2021-2023)

The era was characterized by the maturity of deep learning and the first use of models based on transformers that greatly increased the possibilities of game AI.

- Liu et al. [6] showed that neural methods are superior to the traditional one by undertaking a thorough study of deep learning on PCG. Their results demonstrated that generative AI models such as GANs and VAEs have the potential to produce content of greater diversity and better satisfaction in players across various genres of games.
- The domain expertise remained crucial, such as in the case of a dedicated survey of dungeon generation by Viana et al. [12]. Their work summarized the possible ways of exploiting methods of graph theory and AI planning to design detailed and consistent game worlds.
- Park et al. [1] presented a paradigm shift in architecture of generative agents that are driven by large language models. Figure1 shows the comparison of simulation progress using numerical values.

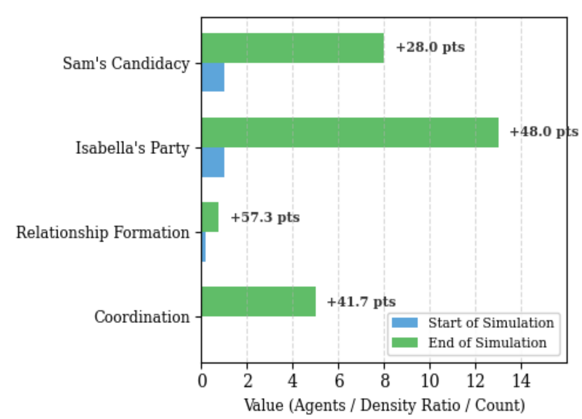

Figure 1 Simulation progress in Generative Agents showing significant growth in information diffusion and social coordination

- Uludağlı and Oğuz [10] was ground-breaking work on generative agents based on the Heavy Language Models to generate interactive simulacra of human behavior with the highest number of citations. Their studies show radical degrees of NPC intelligence, which makes characters have steady personalities, create relationships, participate in complicated social interactions, and show emergent behaviors within a continuing virtual world.

- This era saw a leap in the use of AI in games. Deep learning began to be incorporated into the game AI. New possibilities were unlocked thanks to the transformer and GANs based approaches. Still the domain expertise played a vital role for the quality generation of content. In conclusion these advances pushed games to be ever more interactive, engaging & believable.

D. Generative AI Era (2024-2025)

- The most recent trend in game AI is the prevalence of advanced generative models and emphasis on real-world applications beyond entertainment which are scalable.

- Gupta et al. [7] have outlined systematic review of generative AI and have mapped the rapid expansion of this field. Their survey extended the scope of development of the generative models and their applications in other disciplines, including gaming, using topic modeling.

- Mao et al. [14] topped up the technical aspects of this revolution by providing evidence of the GANs, transformers and diffusion model used to automate the
process of creating high-quality content.

The game AI can be applied across different fields of application, including health and educational contexts (e.g., in serious games), as Tolks et al. [11] explored the applicability of such techniques in that domain and found them useful.

- The 2025 research has continued to expand the range of current game AI. Modern research works into mixed-initiative co-creation between designers and AI [13], examines the role of AI in society through gaming software [19], and adds to the establishment of artificial general intelligence [16].

- Although more modern studies are based on state-of-the-art machine learning, algorithms of procedural content generation are still highly applicable to the generation of structured environments such as dungeons. These classical approaches give a performance and design philosophy that is usually focused on control and predictability.

- Michael [20] offered an exhaustive empirical investigation of four standard dungeon generation methods, namely Binary Space Partitioning (BSP), Depth-First Search (DFS), 2D Delaunay Triangulation (DT) and 3D Delaunay Triangulation (DT) to further elaborate and give a critical performance benchmark in the mentioned problem. The given finding should act as a critical caveat to developers who may wish to use 3D procedural generation, as it highlights the trade-off between spatial complexity and real-time performance

- . This paper connects the early research of Hendrikx et al. [2], who tabulated these algorithmic families, and the current PCGML survey of Summerville et al. [4]. It gives empirical research that in too many practical uses, notably in resource constrained or real time systems, simpler, well-established algorithms may perform better than more complex, data-driven methods on pure performance. This establishes a definite design trade-off between the high efficiency and controllability of traditional algorithms such as DFS and BSP, and the adaptive, high-variety capability of new techniques of PCGML and generative AI.

3. COMPARATIVE ANALYSIS

This part holds understanding in the form of elaborated comparison tables. Evidence-based evaluation was done by a systematic synthesis of the findings that are expressly mentioned in literature studied and evaluated using the following process.:

a) Analysis Methodology

These are the steps involved for the Comparison methodology.

1. Strengths Identification: Benefits, improvements in performance, and benefits described by the authors were reportedly pruned out of individual papers results and discussion sections in a systematic fashion to be able to cover all positive outcomes.
2. Limitations Analysis: The papers Future Directions and Limitations sections are used to assess the limitations of the field and gain insights from it.
3. Evidence Traceability: Every assessment and finding are backed with evidence and proper

citations; the findings are represented in the format of graphs and pictorial representation.

4. Source Attribution: All the sources are provided in the columns so that the researcher can verify the sources and use it for further research, this is done in order to improve the readability of the document and make it more user-friendly.

This approach will ensure that all the assessments expressed in the following tables will have empirical evidence as opposed to subjective interpretation and offer credible grounds on comparative analysis and subsequent research inquiries.

b) Comparison of Methodologies Comprehensively.

Table 1 gives a more detailed summary of all over 15 AI-driven NPC and PCG methods reviewed within this paper listed in terms of research impact and comprises detailed analysis comments regarding each contribution.

TABLE 1: Comprehensive Comparison of All 21 AI-Driven NPC and PCG Methods with Analysis Remarks

S. No.	Ref	Authors	Year	Focus Area	Methodology	Remarks & Key Accomplishments
1	[3]	Yannakakis & Togelius	2011	Experienc e PCG	Player Modeling	Brings about a player-focused procedural generation system that relates psychological models to automatic systems.
2	[2]	Hendrikx et al.	2013	PCG Survey	Comprehensive Analysis	Suggests a Standardized system of classification of PCG methods to different types of games and content.
3	[17]	Ponce & Padilla	2014	Hierarchical RL NPCs	Multi-level RL	Uses its hierarchical reinforcement learning to develop NPCs who can learn in a changing game world.
4	[8]	Al-Emran	2015	Hierarchical RL	Multi-level RL	Multi-level AI architectures in high multi-agent decision-making and planning.
5	[18]	Feng et al.	2016	Autonomo us Learning	ML, Behavior Trees	Learns on its own through the combination of machine learning and behavior trees to allow NPCs to be adaptive and self-learn.
6	[9]	Snodgrass & Ontan˜o´n	2017	Controllable PCG	Markov Chains	Produces game content through controlled probabilistic models that can be inputted by designers in more detail.
7	[4]	Summerville et al.	2018	PCGML Survey	ML Integration	Gives a characterizing survey systematizing PCGML and proposes an elaborate taxonomy of machine learning methods.
8	[5]	Risi & To- gelius	2020	ML Gener- ality	Evolutionary Computation	Positions procedural content generation as an important field of testing and developing artificial intelligence.
9	[15]	Zafar et al.	2020	Search- Based PCG	Evolutionary Al- gorithms	Produces an adaptable and search-oriented process of content creation of various game genres.
10	[6]	Liu et al.	2021	Deep Learn- ing PCG	Neural Networks, GANs	Use deep neural networks such as GANs and VAEs to generate levels of games and images of high quality and diversity.
11	[12]	Viana et al.	2021	Dungeon PCG	Graph Theory, AI Planning	Dungeon generation Systematizes dungeon generation by using graph theory and AI planning algorithms.

12	[1]	Park et al.	2023	NPC Intelli-gence	LLMs, Agent-based Modeling	Trains NPs based on the large language models that are intended to mimic human social behavior and long memory.
13	[10]	Uludag˘lı & Og˘uz	2023	NPC Deci-sion Making	Multiple AI approaches	Makes a comparison of the NPC AI models (reinforcement learning, fuzzy logic, and behavior trees).
14	[7]	Gupta et al.	2024	Generative AI	Topic Modeling, NLP	Systematically and data-driven reviews the current topography of the generative AI techniques.
15	[11]	Tolks et al.	2024	Serious Games	AI Integration	Shows how AI in games can be used to develop serious games that have health-related consequences in serious games with quantifiable therapeutic effects.
16	[14]	Mao et al.	2024	Gener ative AI PCG	GANs, Transformers	Researches on the next-generation content creation with advanced generative models, such as GANs and transformers.
17	[13]	Dutra et al.	2025	Mixed-Initiative PCG	Reinforcement Learning	AI-generated content Presentation of a collaborative PCG model with a combination of human design intuition and AI-generated content.
18	[16]	Raman et al.	2025	AGI Analy-sis	Policy, Ethics	Evaluates the ethical implications and social outcomes of the progressive AI in the gaming industry.
19	[19]	Kumar et al.	2025	AI Impact Analysis	Societal Studies	Discusses the overall implications of AI technologies on the process of game-making and the communities of players.
20	[20]	Michael	2025	Algorithm for Dungeon Generation	controlled, empirical performance benchmark	Integrates Quantitative and Qualitative analysis to analyze different algorithms.

c) Quantitative Comparative Analysis

1. **Constrained Sampling Approaches Performance Analysis:**

In order to offer further information about the empirical workings of constrained procedural content generation techniques, we examine the experimental outcomes of the seminal publication by Snodgrass and Ontanos concerning constrained multi-dimensional Markov chain sampling [9]. They compared three constraint enforcement methods in two traditional video game fields and found that there are significant trade-offs between constraint satisfaction and computational efficiency.

Figure 2 is a heatmap-like performance matrix that visualizes the main experimental results of Kid Icarus and Super Mario.

Figure 2 Algorithmic Performance: Green indicates optimal performance; Red indicates high cost/failure

2. **The most important lessons learned in Constrained Sampling Performance:**

Violation Location Resampling (VLR) is the most balanced of them, with high rates of satisfaction (75-100%) and moderate computational cost (23-176 attempts)

Incremental Sampling (IS) works best in the context of more basic sets of constraints, where it satisfies its objective to near perfection with only nearly minimal effort (21.02 in the case of Super Mario Bros.).

Generate and Test (G + T) is high-satisfaction and when using hard constraints is very costly (435.96 attempts).

The algorithms demonstrated high effectiveness in Kid Icarus, in which baseline MdMC sampling was entirely unsatisfactory (0% satisfaction).

This experimental study highlights the importance of the choice of algorithm depending on the complexity of constraints and the nature of the domain, which is of practical value to developers who use constrained PCG systems.

3. Empirical Traditional Dungeon Generation Algorithms Performance Analysis:

In order to give proper and solid evidence of the performance trade-offs in the traditional procedural content generation algorithms, we examine the experimental findings of a detailed study by Michael [20] of comparing four dungeon generation algorithms: Depth-First Search (DFS), Binary Space Partitioning (BSP), 2D Delaunay triangulation (DT 2D) and 3D Delaunay triangulation (DT 3D).

Figure 3: Performances Comparison of Dungeon Generation Algorithms [20]. Radar chart is used to show the relative performance of four algorithms against three important metrics. DFS is characterized by even efficiency and DT 3D by extreme resource requirements.

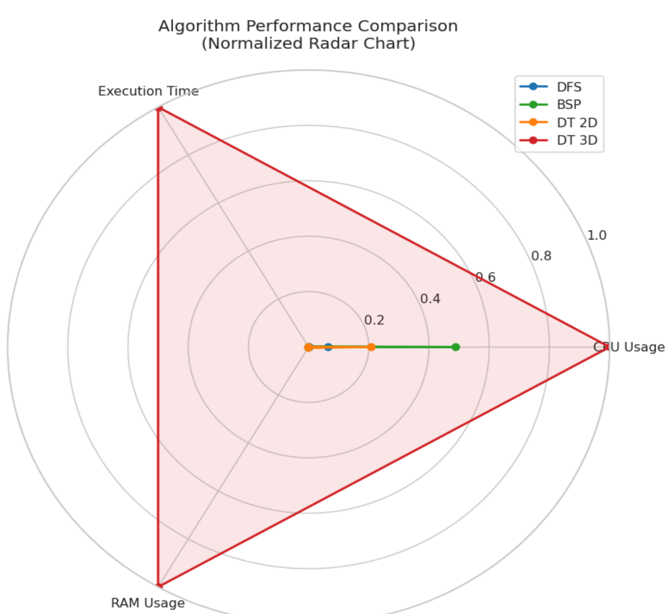

Figure 3 Algorithm Performance Radar Chart comparing different algorithms performances

The most important Quantitative Findings of the work of Michael [20] are shown in Figure 4:

DFS was highly efficient, DT 2D was moderate in its usage of resources but provided a wider variety of room layouts. BSP had greater computational requirements although it has a theoretically comparable computation complexity to DT 2D. DT 3D had terribly poor performance, so it was not useful in large scale generation.

Figure 4 shows these findings in the form of a Bar Chart.

Algorithmic Trade-offs and Applications:

The discrepancy in performance can be explained by the complexity of the algorithm and peculiarities of its implementation. DFS has O(V+E) complexity and is an efficient way of creating maze-like structure with minimal layout diversity.

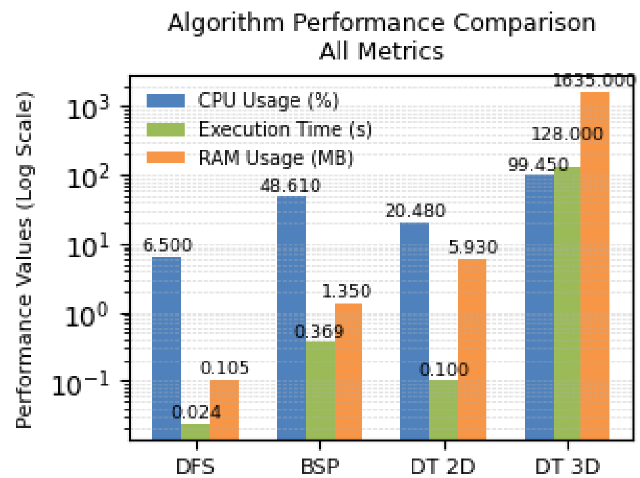

Figure 4 Algorithmic Performance Comparison between different algorithms on different constraints

But have a practical complexity difference of overheads in the generation of corridors and pathfinding. The complexity of DT 3D at O (n 2) with advanced staircase pathfinding and 3D spatial reasoning leads to the growth of resources.

c) Techniques Assessment

Fig 5 shows the Overview of the AI Techniques used and a diagrammatic representation of it.

4. FINDINGS AND FUTURE DIRECTIONS

The academic environment in AI-generated game development has seen a significant change during the last ten years. An analysis of its contributions shows that; besides the methodological growth, a new definition of the nature of intelligent game content and behavior is also emerging. However, instead of giving individual works in isolated summations, the section will synthesize cross cutting observations based on the surveyed literature, with an overall agenda on the unresolved constraints and new research priorities.

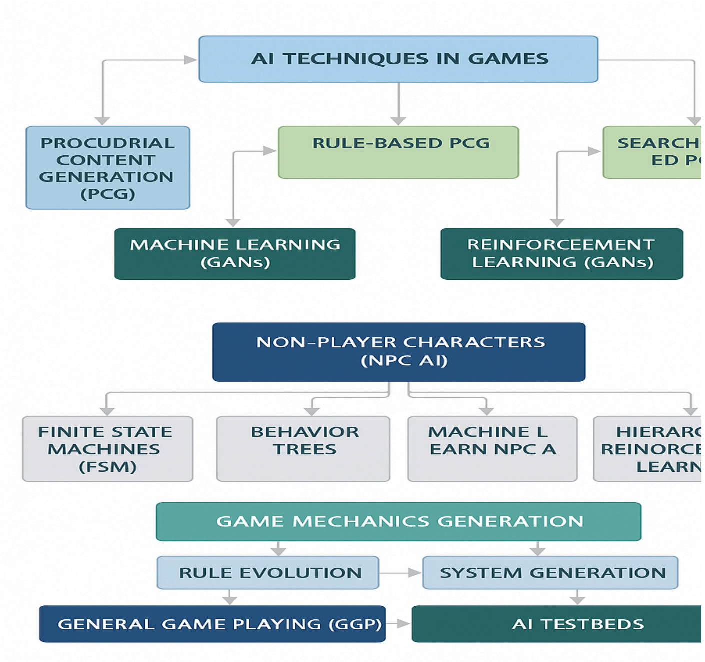

Figure 5 AI Techniques used in Games

AI was expanded in games further by the introduction of agent-based architectures, which had memory and reasoning layers and self-refinement procedures as shown by Park et al. [1].

1. The Changing Paradigms of AI-Enhanced Game Design.

One change can be discerned is in the perception of content generation systems and game intelligence . Initial content generation (PCG) studies concentrated greatly on rule-based content pipelines, which were hand-crafted and based on designer-coded heuristics to provide variety and control [2].

2. Hybrid Synthesis Insight:

The area is no longer on a linear "rules - ML - neural models" path. Rather, it is moving towards hybrid intelligences which are a result of a combination of procedural constraint, learned priors, and independent agent reasoning.Diversifying Methodologically and the Implications.

With the integration of neural generative models, one was able to create more context-sensitive artefacts, such as playable levels all the way up to narrative constructs [4], [6]. Simultaneously, studies of PCG are no longer limited to content generation, and this approach is currently also acknowledged as an instrument to investigate model generalization and transferability, especially in reinforcement learning contexts [5].

4. Constant Limitations Baffling Field Development

Although there has been a great improvement, a number of structural limitations are still evident throughout the literature.

These limitations are indicated by Table 2.

Table 2 Constant Limitations Baffling Field Development

Limitation	Description	Impact
Modern AI pipelines are having computational costs	Deep generative models and reflective agent architectures are very expensive in terms of processing [1]	Disallows real-time games and limited resources
Disjointed appraisal norms	A quality PCG, diversity, and player-centered value are measured inconsistently in the literature [2],[4]	Impairs comparability, reproducibility and accumulations of knowledge
The lack of data and field specificity	ML-based PCG usually requires organized data, which is not available to new genres of games [6]	Imposes obstacles to innovation in new or non-mainstream forms of games
Lack of cross system flexibility	The majority of systems have limited application outside of the field of design [4], [5]	Hinders development of generalizable AI in games

5. Future Research Agenda

Synthesized trends and gaps lead to the following directions of research: **Construction of Multimodal, Contextually Grounded AI Systems**, **Change to more system centric assessment**, **Standardization of Evaluation and Common Benchmarks, Scalable and Transferable AI Design Process**, **AI Design based on Ethics, Responsibility and the Personality of the Player.**

Final Perspective: The history of AI was using traditional algorithms to improve model efficiency, but now social and economic behavior are also taken into account. These increase the complexity of the system and allow it to simulate human behavior better.

6. CONCLUSION

This review has been the documentation of the last 15 years or so in the field of Artificial Intelligence in games, particularly NPC Intelligence and PCG. The results indicate that there has been a significant change from the rule-based and hand-crafted system to the data-based one such as a combination of deep learning and large language models of generative agents. In the literature , one similar observation was the hybrid systems between the constraint of procedures and learned models.Empirical studies show that modern advances in AI show better realism and variety but, the classical algorithms like DFS, BSP and others are more predictable, efficient and have good performance in real-time. This points to a critical trade-off between controllability and flexibility in generation.

To sum it up this discussion indicates a need for standardized evaluation, scalable architecture and informed design that is ethical implying an equal mix of classic algorithms and sophisticated generation models in future game AI systems.

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